

Citadel

Masum Billal

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Citadel

The Knowledge Compilation Layer for Enterprise AI

The foundational layer every reliable AI system needs

The Architectural Void

Enterprise knowledge remains trapped inside human-centric visual formats (PDFs, spreadsheets, reports). Existing systems optimize for storing and retrieving artifacts, not executing over knowledge.

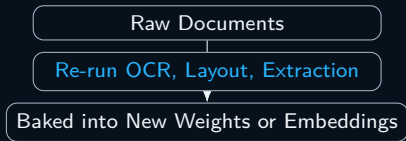
The Universal Primitive

Citadel introduces the missing system layer: compiling unstructured syntax into a persistent, high-fidelity computational substrate so that AI applications can seamlessly execute native data workflows.

Every AI System Re-Pays the Same Hidden Cost

Every AI system couples enterprise data to one model's representation — its weights, or its embedding space. Both approaches skip the same intermediary step, and both pay for it the same way, on every model swap.

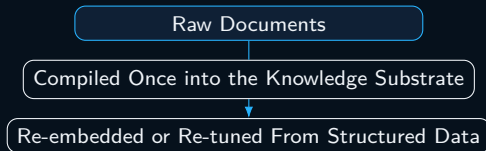
Without a Knowledge Layer



Full reingestion cost, paid again on every model swap.

Models iterate weekly. Applications pivot monthly.

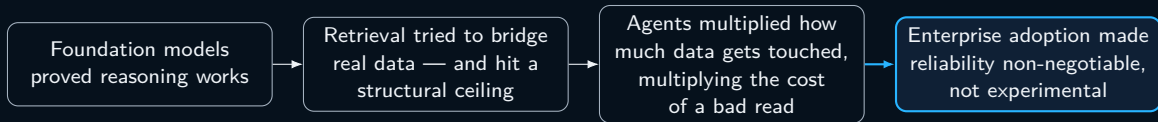
With Citadel



Extraction cost paid once. A model swap touches only the substrate.

Infrastructure persists for decades.

Why Now



The missing layer was survivable when AI was experimental. It stopped being optional the moment reliability became the product.

The Missing Systems Tier for Enterprise Intelligence

Citadel serves as the stable computational foundation layer, transforming raw inputs into invariant state.

Downstream Enterprise AI Workflows & Custom Interfaces

Reasoning Layer (LLMs)

Probabilistic reasoning engine · Not an authoritative database

Knowledge Substrate Layer (Citadel)

Enterprise Reality

Multi-format documentation, structural data, multi-modal files

Raw data contains fragments.

**Citadel acts as the
deterministic state engine.**

Models execute reasoning over
it.

The Foundational Layer Every AI Company Will Require

Any AI product whose value depends on reading, structuring, or reasoning over real documents faces the same dependency — and a competitor built on a solved version of it carries a structural, compounding advantage.

Not a Feature. A Survival Dependency.

- ▶ An extraction error is not local — every downstream layer built on top of a wrong read inherits it.
- ▶ As the AI market matures past novelty into reliability, the ingestion layer stops being invisible and becomes the differentiator.

▶ The gap compounds. it does not

Who Already Depends On It

- ▶ **New AI companies**, at the build-vs-buy decision for ingestion — the same default shift that made hand-rolled payments infrastructure disappear once Stripe existed.
- ▶ Enterprises building internal AI and agents over their own document corpora — direct buyers, not only vendors.

Not a Claim. A Measured Result.

One messy spreadsheet, four unrelated datasets on one sheet. Asked: how many tables, how many rows.

so how many tables did you say the excel file had and how many rows in each one?

From this workbook, I identified 3 logical tables on the single worksheet.

Table	Approximate Rows	Number of Rows	Description
Table 1	3-18	18 rows	Black-Scholes option pricing model (lookup table, inputs, outputs, calculations)
Table 2	11-24	3 rows	Dataset metadata (World Development Indicators source information)
Table 3	25-62190	62,198 rows	World Development Indicators dataset

Total

- Worksheets: 1
- Logical tables: 3
- Total rows across all tables: 62,185 rows

Notes

The third table is by far the largest:

- Rows: 62,166
- Columns: roughly 70 (Country Name, Country Code, Indicator Name, Indicator Code, followed by yearly columns from 1960 through 2025)

ChatGPT, given the raw file

3 tables detected

Missed two entire datasets it never reported at all.

“62,166 rows” claimed for one table

The whole workbook has 5,785 rows.

Not close. A model reasoning over raw bytes cannot see structure that was never extracted.

Not a Model Problem. A Missing Capability.

Same file, given to a model that wrote and ran its own verification code — real execution, not guessing.

2. Detailed Data Sections & Row Counts

Sheet 1: `Sheet1`

`Sheet1` contains 5,785 total occupied rows spanning several distinct data tables/blocks:

- Option Distribution Calculations (Rows 3-5): 3 rows
- Option Inputs & Parameters (Rows 7-12): 6 rows
- Calculated Financial Values (Rows 14-18): 5 rows
- Data Source Metadata (Rows 19-23): 5 rows
- Main Dataset (World Development Indicators) (Rows 25-5785):
 - Header row: Row 25
 - Data rows: Rows 26-5784 (5,759 rows)
 - Total row: Row 5785 (Total Result)

Sheet 2: `Pivot Table_Sheet1_1`

This sheet contains a single Pivot Table with 181 total rows:

- Header Row: Row 1 (Country, Year, Sum - Total)
- Data Rows: Rows 2-180 (179 country summary rows)
- Summary Row: Row 181 (Total Result: 11,784)

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+ Ask Gemini

Flash



Gemini, after running its own code

Got the raw sheet size exactly right

5,785 rows, 2 worksheets — verified by code, not hallucinated.

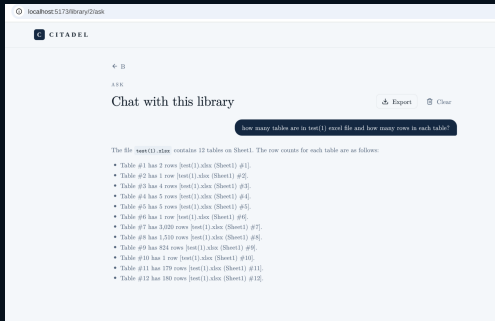
Then called rows 25–5785 “one dataset”

A sheet reads as one flat table from its first header — no concept of a second table starting underneath.

It verified its own numbers. It still cannot see a table boundary.

What Citadel Actually Returns

Same file, same question — answered from stored structure, not raw bytes.



Citadel, given the same file

12 tables detected, correctly typed

Every stacked region found and split — nothing merged,
nothing missed.

5,733 rows, verified against source

Every count checked cell-for-cell against the original file.

Not a better guess. The structure was already there.

A Harder Test, Not an Easier One

ChatGPT and Gemini were handed this one spreadsheet, alone, with nothing else to search. Citadel answered the same question from a library of roughly 30 files — and neither tool can even accept that scale in the first place.

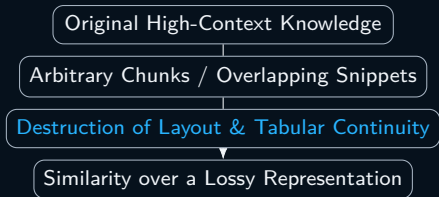
	ChatGPT	Gemini	Citadel
Files given	1, in isolation	1, in isolation	~30, had to find it
Files/size supported	hard platform limits	hard platform limits	unlimited

Citadel had the harder job — find it, then read it, at a scale the others cannot even accept — and still got every number right.

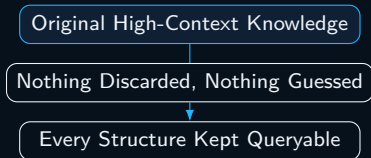
Retrieval Cannot Recover What Structurally Disappeared

If a representation discards structure, nothing downstream recovers it. Every answer must trace back to exactly where it came from — nothing flattened, nothing untraceable.

The Retrieval Failure Mode



The Citadel Paradigm



Enterprise First, Infrastructure Next

Phase One: Direct to Enterprise

- ▶ **Legal** — multi-contract structural risk analysis.
- ▶ **Financial** — cross-document reconciliation audits.
- ▶ **Real Estate** — title, lease, and property-record intelligence.
- ▶ **Regulatory** compliance and due diligence.

Phase Two: Infrastructure

Once proven directly with enterprise buyers, the same substrate opens as the ingestion and retrieval layer other AI companies build on — sold as platform, not a point solution.

One layer proven with the enterprises, then opened to everyone.

Two Tiers, Priced on Usage

Structure

Lossless ingestion into SQL-queryable tables. No retrieval, no Q&A — the raw substrate itself, billed on volume ingested and retained.

Search

Everything in Structure, plus retrieval and natural-language Q&A over it — billed on volume plus queries executed.

Annual enterprise contracts in Phase One — metered API access once the substrate opens as infrastructure in Phase Two.

Priced on what it does for you, not seats or licenses.

The Ask

Raising **\$3,000,000** for **10%** of the company (\$30M post-money) — priced for a working system, not a concept.

What This Round Is For

- ▶ Founder, full-time.
- ▶ First 1-3 hires beyond the founder.
- ▶ Compute and infrastructure to deploy into real cloud infra and onboard real users.
- ▶ Legal, compliance, and IP setup for enterprise-grade deployments.

What It Buys

- ▶ 18–24 months of runway.
- ▶ The technical barrier is already crossed — this capital converts a working system into paying design partners.
- ▶ Target: signed pilots and repeatable proof this isn't just an impressive demo.
- ▶ Hardware, cloud, human and all other resources

Team

Masum Billal

Founder

Solo-built: ingestion pipeline, query engine, and everything else this deck shows.
Every result in this deck is measured against a working system, not a mockup.